LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 2 Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts · Week 1 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each. Do Day 1 on the day you study the chapter, then Day 2 and Day 3 on the next two days.
2. Write your answers in your notebook, then check them with the key the same day. Copy every mistake into your error log.
3. Learn the formula and one use of each salt: baking soda, washing soda, bleaching powder and Plaster of Paris.

DAY 1: ACIDS, BASES AND INDICATORS

Q1. Which of the following is an olfactory indicator?

(a) litmus (b) clove oil
(c) phenolphthalein (d) methyl orange

Q2. Which acid is present in tamarind?

(a) citric acid (b) lactic acid
(c) oxalic acid (d) tartaric acid

Q3. In water, acids produce:

(a) OH^- ions (b) H^+ ions (as H_3O^+)
(c) Na^+ ions (d) Cl^- ions only

Q4. A base that dissolves in water is called:

(a) a salt (b) an indicator
(c) an oxide (d) an alkali

Q5. Name the acid present in: (i) vinegar (ii) curd (iii) an ant sting (iv) lemon.

Q6. HCl and H_2SO_4 show acidic properties in water, but solutions of glucose and alcohol do not, although they also contain hydrogen. Why?

Q7. Zinc reacts with sodium hydroxide solution on warming. Write the balanced equation and name the gas formed.

Q8. You have three test tubes. One contains distilled water, one an acid and one a base. You have only red litmus paper. How will you identify each one?

DAY 2: REACTIONS AND PH

Q9. Metal oxides are generally:

(a) acidic (b) neutral
(c) basic (d) always amphoteric

Q10. The pH of the gastric juice in our stomach is about:

(a) 2 (b) 7
(c) 9 (d) 12

Q11. Solutions of equal concentration are made of each substance below. Which has the lowest pH?

(a) HCl (b) NaOH
(c) NaCl (d) CH_3COOH

Q12. Write the balanced equations for the reactions of dilute hydrochloric acid with (i) copper(II) oxide and (ii) magnesium hydroxide. What do both reactions have in common?

Q13. Why does an ant sting cause pain and a burning feeling? What can be applied to the skin for relief, and why does it work?

Q14. Fresh milk has a pH of about 6.
(a) How does the pH change as it turns into curd? Why?
(b) A milkman adds a very small amount of baking soda to fresh milk. Why?
(c) Why does this milk take longer to set as curd?

Q15. Salt solutions can be acidic, basic or neutral. Classify each with a reason: (i) NaCl (ii) Na_2CO_3 (iii) NH_4Cl.

DAY 3: SALTS, MIXED PRACTICE AND CHALLENGE

Q16. **Assertion (A):** Plaster of Paris sets into a hard mass when it is mixed with water.
Reason (R): Plaster of Paris contains no water of crystallisation.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q17. Bleaching powder is made by passing chlorine gas over:
(a) dry slaked lime (b) quicklime
(c) limestone (d) gypsum

Q18. Riya got a turmeric (haldi) stain on her white shirt. When she scrubbed it with soap, the stain turned reddish-brown. After she rinsed it with plenty of water, it turned yellow again.

(i) Why did the stain turn reddish-brown?

(ii) Why did it turn yellow again after rinsing?

(iii) Name one more natural indicator and one synthetic indicator.

Q19. (a) A few drops of phenolphthalein are added to sodium hydroxide solution. Dilute hydrochloric acid is then added drop by drop. Describe the colour changes, explain them and write the balanced equation.
(b) What will you see if a few drops of NaOH are added again?
(c) Name the salt formed. Is its solution acidic, basic or neutral?

Q20. A calcium compound A reacts with dilute hydrochloric acid with effervescence. The gas given off puts out a burning candle. One of the products is calcium chloride. Identify A and the gas, and write the balanced equation.

~ Week 1 done after this one. Ch 1 and Ch 2 tests next. You've got this! ~